

Exact methods

Gomory cuts

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Optimization: principles and algorithms

Gomory cuts

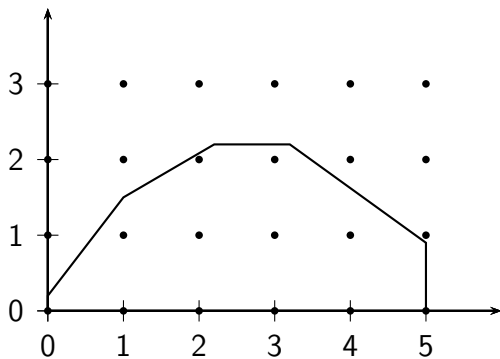
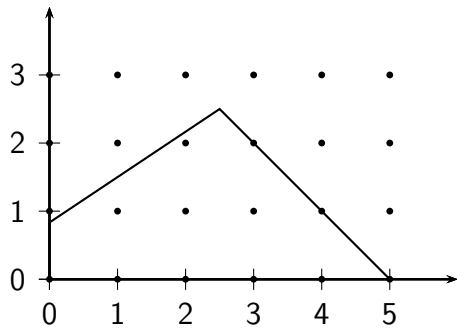
*It is possible to shrink the constraint polyhedron so that its vertices are integer
In that case, the simplex algorithm will solve the problem.*

In practice, additional constraints must be added.

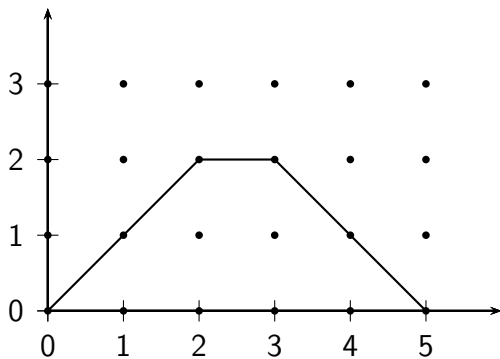
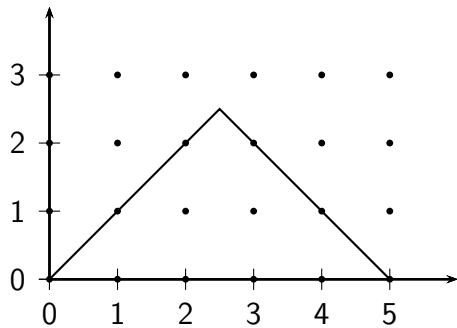
*They must shrink the feasible set of the relaxation, but not of the original
discrete optimization problem.*

*These constraints are called “valid inequalities” when they are designed by the
modeler, and “cutting planes” or “cuts” when they are generated by an
algorithm.*

Feasible set of integer optimization



Feasible set of integer optimization



Simplex tableau

$B^{-1}A$	$B^{-1}b$
$c^T - c_B^T B^{-1}A$	$-c_B^T B^{-1}b$

We have an optimal tableau for the relaxed problem.

Upper part: a modified version of the constraints

Decompose the upper part into basic and non basic variables

$$B^{-1}Ax = B^{-1}b,$$

$$B^{-1}Bx_B + B^{-1}Nx_N = B^{-1}b,$$

$$x_B + B^{-1}Nx_N = B^{-1}b.$$

Basic variables are first, so that $x_i = (x_B)_i$, $i \leq m$. Consider row i and call α_{ij} the element of the tableau

$$x_i + \sum_{j \text{ non basic}} \alpha_{ij} x_j = (x_B^*)_i.$$

Rounding the constraint

$$x_i + \sum_{j \text{ non basic}} \alpha_{ij} x_j = (x_B^*)_i.$$

x is feasible, it is non negative.

Rounding down all coefficients α_{ij} , the value cannot increase

Valid inequality for all feasible solutions of the relaxation

$$x_i + \sum_{j \text{ non basic}} \lfloor \alpha_{ij} \rfloor x_j \leq x_i + \sum_{j \text{ non basic}} \alpha_{ij} x_j = (x_B^*)_i.$$

Rounding again the constraint

$$x_i + \sum_{j \text{ non basic}} \lfloor \alpha_{ij} \rfloor x_j \leq (x_B^*)_i.$$

$$\left\lfloor x_i + \sum_{j \text{ non basic}} \lfloor \alpha_{ij} \rfloor x_j \right\rfloor \leq \lfloor (x_B^*)_i \rfloor.$$

x is integer

$$\left\lfloor x_i + \sum_{j \text{ non basic}} \lfloor \alpha_{ij} \rfloor x_j \right\rfloor \leq \lfloor (x_B^*)_i \rfloor.$$

$$\left\lfloor x_i + \sum_{j \text{ non basic}} \lfloor \alpha_{ij} \rfloor x_j \right\rfloor = x_i + \sum_{j \text{ non basic}} \lfloor \alpha_{ij} \rfloor x_j$$

$$x_i + \sum_{j \text{ non basic}} \lfloor \alpha_{ij} \rfloor x_j \leq \lfloor (x_B^*)_i \rfloor,$$

Valid inequality for all feasible solution of the integer problem: it is called a Gomory cut

Solution of the relaxation

$$x_i + \sum_{j \text{ non basic}} \lfloor \alpha_{ij} \rfloor x_j \leq \lfloor (x_B^*)_i \rfloor.$$

If x^* is solution of the relaxation:

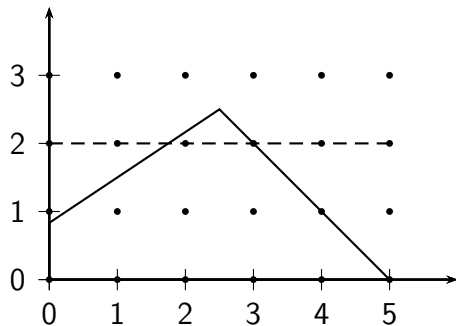
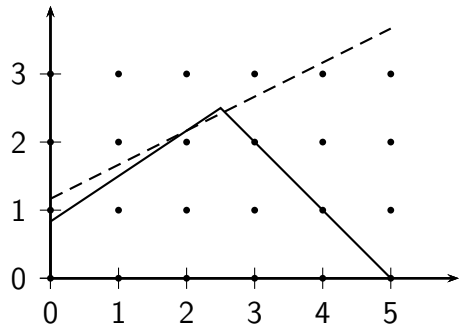
All non basic variables are 0

$$x_i^* \leq \lfloor (x_B^*)_i \rfloor = \lfloor x_i^* \rfloor,$$

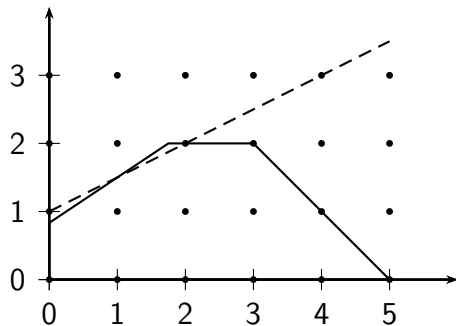
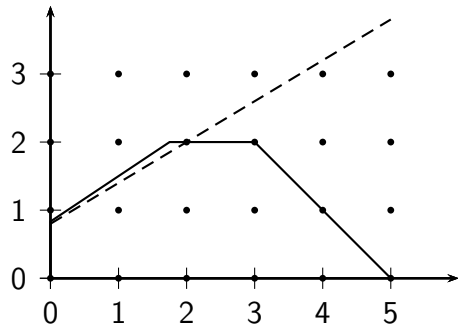
This is verified only if x_i^ is integer*

Therefore, if i is chosen such that x_i is fractional, it generates a cut that excludes x^ .*

Examples



Examples



Summary

The branch & bound was cutting the constraint polyhedron along the axes of coordinates.

Gomory cuts exploit the structure of the simplex tableau to generate valid inequalities, that exclude (if well chosen) the optimal solution of the relaxation. It means that, at each iteration, the polyhedron is made strictly smaller. Eventually, all the vertices will be integer.